

Multiple Interface Inheritance in Java

Basic Information

Assignment: Assignment 2

Program: MultipleInterfaceInheritanceDemo.java

Topic: Interfaces — Different Types of Implementations of Interfaces

Objective / Aim

To demonstrate an interface extending multiple interfaces and a class implementing the child interface.

System Requirements

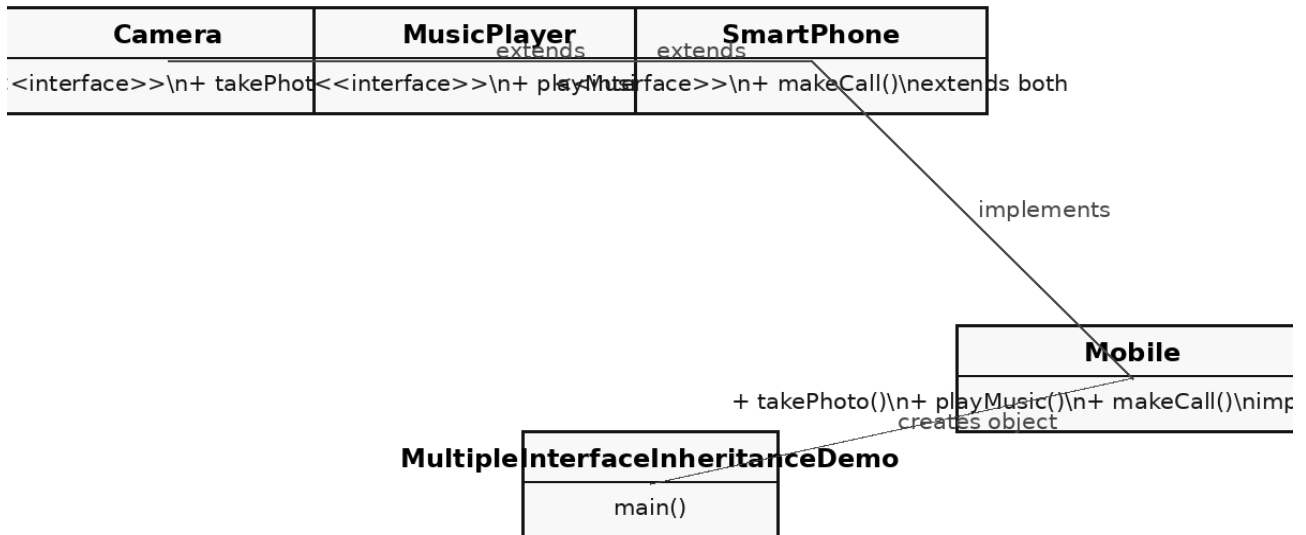
Hardware: Computer/Laptop, minimum 4 GB RAM, keyboard and mouse.

Software: Java Development Kit (JDK), Java compiler, Java Runtime Environment (JRE), and any Java-supported IDE or terminal.

Algorithm / Logic Description

1. Start the program.
2. Create Camera with takePhoto().
3. Create MusicPlayer with playMusic().
4. Create SmartPhone that extends both Camera and MusicPlayer and declares makeCall().
5. Create Mobile and implement SmartPhone.
6. Define takePhoto(), playMusic(), and makeCall() in Mobile.
7. Create a Mobile object in main().
8. Call all three methods using the object.
9. Stop the program.

UML Class Diagram



Source Code

```

// First interface
interface Camera {

    void takePhoto();
}

// Second interface
interface MusicPlayer {

    void playMusic();
}

// Interface extends multiple interfaces
interface SmartPhone extends Camera, MusicPlayer {

    void makeCall();
}

// Class implements the child interface
class Mobile implements SmartPhone {

    public void takePhoto() {
        System.out.println("Taking photo");
    }

    public void playMusic() {
        System.out.println("Playing music");
    }

    public void makeCall() {
        System.out.println("Making call");
    }
}

public class MultipleInterfaceInheritanceDemo {
    public static void main(String[] args) {

        Mobile mobile = new Mobile();

        mobile.takePhoto();
        mobile.playMusic();
        mobile.makeCall();
    }
}
  
```

```
}
```

Compilation & Execution Commands

Save as: MultipleInterfaceInheritanceDemo.java

Compile: javac MultipleInterfaceInheritanceDemo.java

Execute: java MultipleInterfaceInheritanceDemo

Output Screen

```
C:\Java\Assignment2> java Program
Taking photo
Playing music
Making call
```

Result

Thus, the multiple interface inheritance program was implemented successfully using the interface concepts demonstrated in the source code.